Chapter 10: Photosynthesis

Complete Master Reference Edition • Fused Curriculum (Introduction, The Light Reaction, The Calvin Cycle, Lecture Slides & Practice Questions)

Course: BSC2010C General Biology I

Institution: University of Central Florida

Source: Canvas Modules, Instructor Slides & Active Learning #3

Section 10.1: Content Introduction & Chloroplast Foundations

Biological energetics, chloroplast anatomy, and light absorption properties.

Chapter 10 Content Introduction

Learning Objectives

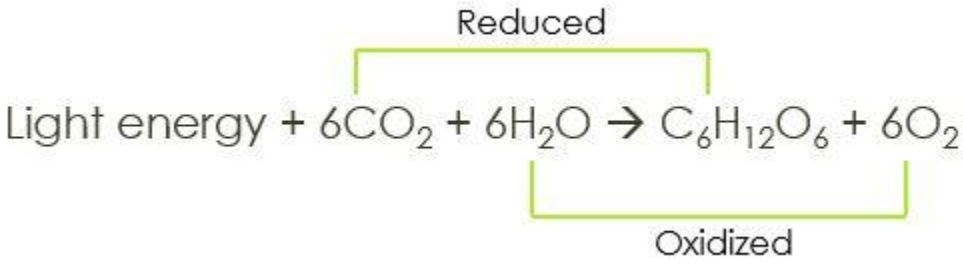
- Describe chloroplast structure
- Compare/Contrast cellular respiration equation with photosynthesis equation
- Explain pigment absorption/reflection
- Differentiate between the Light Reaction and Calvin Cycle
- Compare the function of Photosystem II and Photosystem I
- Explain production of ATP and NADPH by Light Reaction
- Explain how carbon fixation leads to the creation of G3P
- Describe how RuBP is regenerated

Chapter 10: Photosynthesis

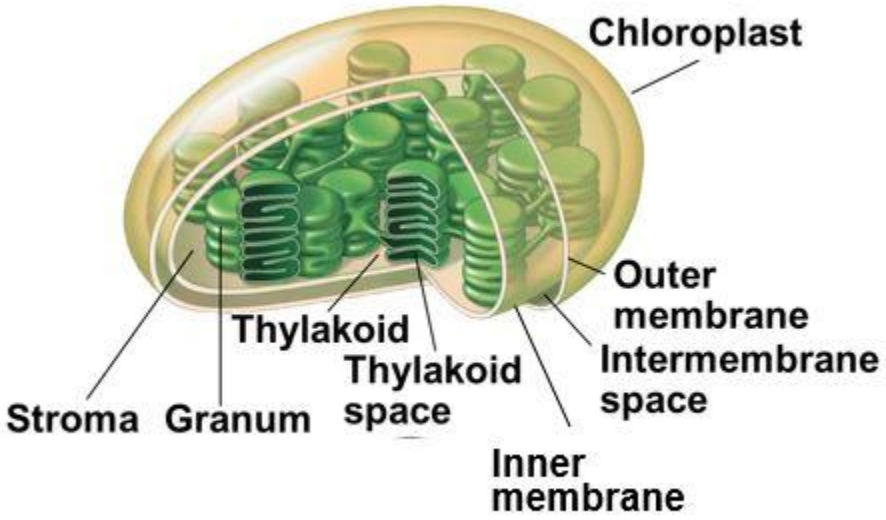
Did you know that life on Earth is solar powered!? Much like we humans can harness the energy in sunlight to warm our pools, charge our cell phones, or even power our houses, plants harness the energy in sunlight (plus CO_2 and water) to create sugar, this is called **photosynthesis**. This sugar is then broken down during cellular respiration to create ATP that the plant's cells use to carry out all the functions of life. Since plants make their own food, we call them **autotrophs,** specifically they are **photoautotrophs** since they use light energy. Organisms that ingest food from an outside source are called **heterotrophs**. Plants not only feed themselves, they feed the world. Heterotrophs consume plants and/or they consume animals that consumed plants.



Photosynthesis is a redox reaction in which CO_2 is reduced to $C_6H_{12}O_6$ and H_2O is oxidized to O_2. Photosynthesis is anabolic (it builds sugar) which means that it is endergonic and requires energy.



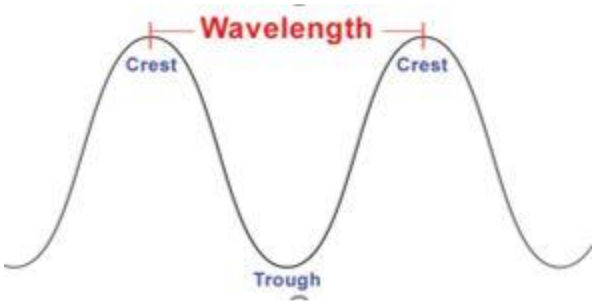
Photosynthesis takes place in special organelles called chloroplasts. Chloroplasts are present in large quantities in the leaves of plants. Let's take a look at the structure of chloroplasts:



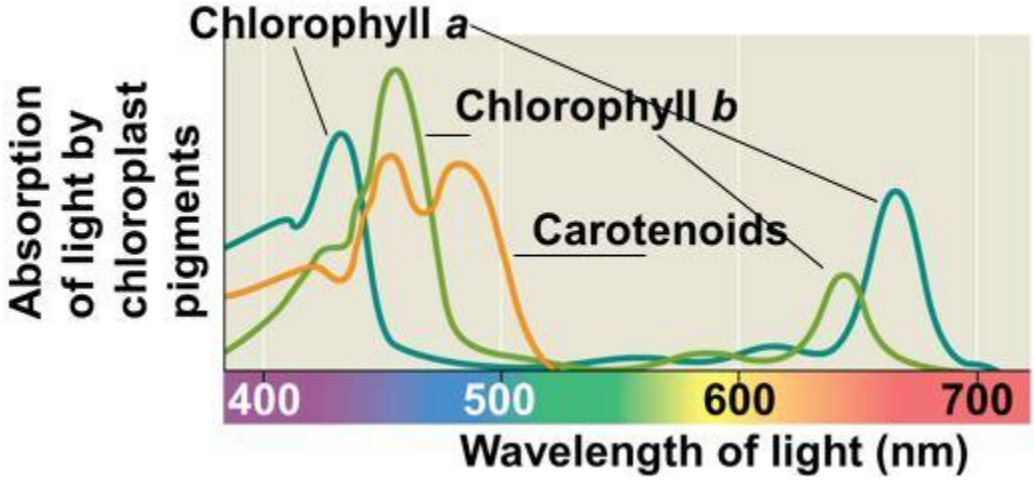
Much like mitochondria, chloroplast have two membranes; an outer membrane and an inner membrane. The space between these two membranes is called the intermembrane space. The space inside the inner membrane is called the stroma (this is analogous to the matrix of a mitochondria). Inside the stroma are thylakoids; they are a membranous system that are stacked in columns called granum. Inside a thylakoid is the thylakoid space, the membrane of the thylakoid separates the thylakoid space from the stoma.

Interactive Course Module: Chloroplast Structure widget

Sunlight is solar energy in the form of electromagnetic waves; these are photons of light that are traveling in a wave. The space between each crest of the wave is called the wavelength.



Visible light falls in the wavelength range of 380-750 nm. These are the colors of the rainbow: red, orange, yellow, green, blue, indigo, and violet (ROY G BIV). Photosynthesis uses these same wavelengths of light. Plants contain several different photosynthetic pigments: Chlorophyll a, Chlorophyll b, and carotenoids. Each pigment absorbs light of a different wavelength. Each pigment reflects light that is not used for photosynthesis (i.e., green). The following graphs shows which wavelengths (and colors) are absorbed by Chlorophyll a, Chlorophyll b, and carotenoids. When you read the graph, pay attention to the colors indicated by the wavelengths at the bottom of the page, not the colors of the individual lines - those are colored based on the color of Chlorophyll a (dark green), Chlorophyll b (light green), and carotenoids (orange). I also want you to notice which color is NOT absorbed.



Interactive Course Module: Light Wavelengths widget

CHAPTER READING

Chapter 10 Textbook Reading: The Light Reaction, The Calvin Cycle

CHAPTER LINKS

Chapter 10 PowerPoint

Chapter 10 Post-Quiz

Chapter 10 Simple Practice Problems

Exam 3 Study Guide

Section 10.2: The Light Reaction (Light-Dependent Phase)

Electromagnetic spectrum, photosystems II & I, water photolysis, and photophosphorylation chemiosmosis.

Chapter 10 The Light Reaction

Learning Objectives

By the end of this section, you will be able to do the following:

- Explain how plants absorb energy from sunlight
- Describe short and long wavelengths of light
- Describe how and where photosynthesis takes place within a plant

How can light energy be used to make food? When a person turns on a lamp, electrical energy becomes light energy. Like all other forms of kinetic energy, light can travel, change form, and be harnessed to do work. In the case of photosynthesis, light energy is converted into chemical energy, which photoautotrophs use to build basic carbohydrate molecules (Figure 8.9). However, autotrophs only use a few specific wavelengths of sunlight.



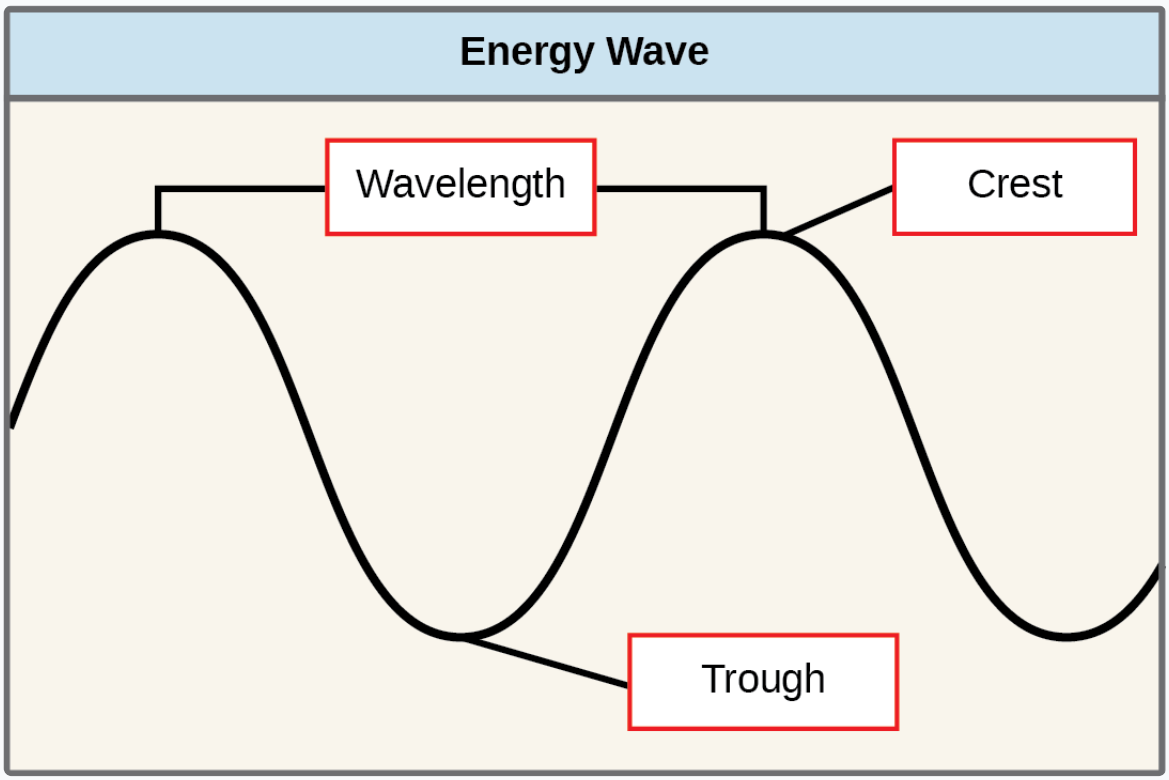
A photo shows the silhouette of a grassy plant against the sun at sunset.

Figure 8.9 Photoautotrophs can capture visible light energy in specific wavelengths from the sun, converting it into the chemical energy used to build food molecules. (credit: Gerry Atwell)

What Is Light Energy?

The sun emits an enormous amount of electromagnetic radiation (solar energy in a spectrum from very short gamma rays to very long radio waves). Humans can see only a tiny fraction of this energy, which we refer to as "visible light." The manner in which solar energy travels is described as waves. Scientists can determine the amount of energy of a wave by measuring its wavelength (shorter wavelengths are more powerful than longer

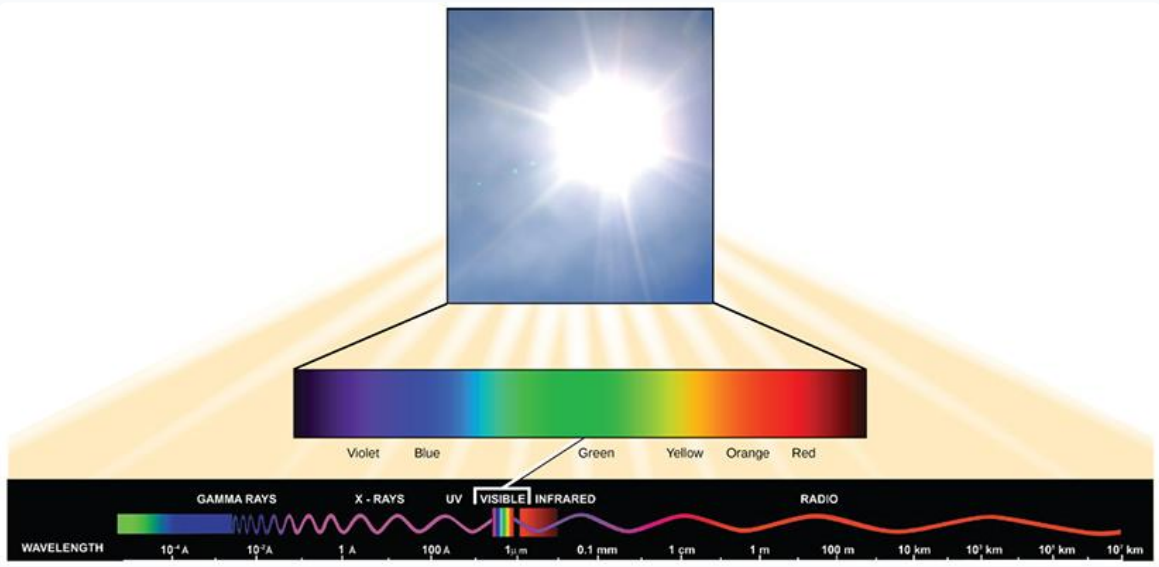
wavelengths)—the distance between consecutive crest points of a wave. Therefore, a single wave is measured from two consecutive points, such as from crest to crest or from trough to trough (Figure 8.10).



The illustration shows two waves. The distance between the crests (or troughs) is the wavelength. The crest is the upper portion of the wave, the trough is the lower portion of the wave.

Figure 8.10 The wavelength of a single wave is the distance between two consecutive points of similar position (two crests or two troughs) along the wave.

Visible light constitutes only one of many types of electromagnetic radiation emitted from the sun and other stars. Scientists differentiate the various types of radiant energy from the sun within the electromagnetic spectrum. The electromagnetic spectrum is the range of all possible frequencies of radiation (Figure 8.11). The difference between wavelengths relates to the amount of energy carried by them.



The illustration lists the types of electromagnetic radiation in order of increasing wavelength. These include gamma rays, X-rays, ultraviolet, visible, infrared, and radio. Gamma rays have a very short wavelength, on the order of one thous...

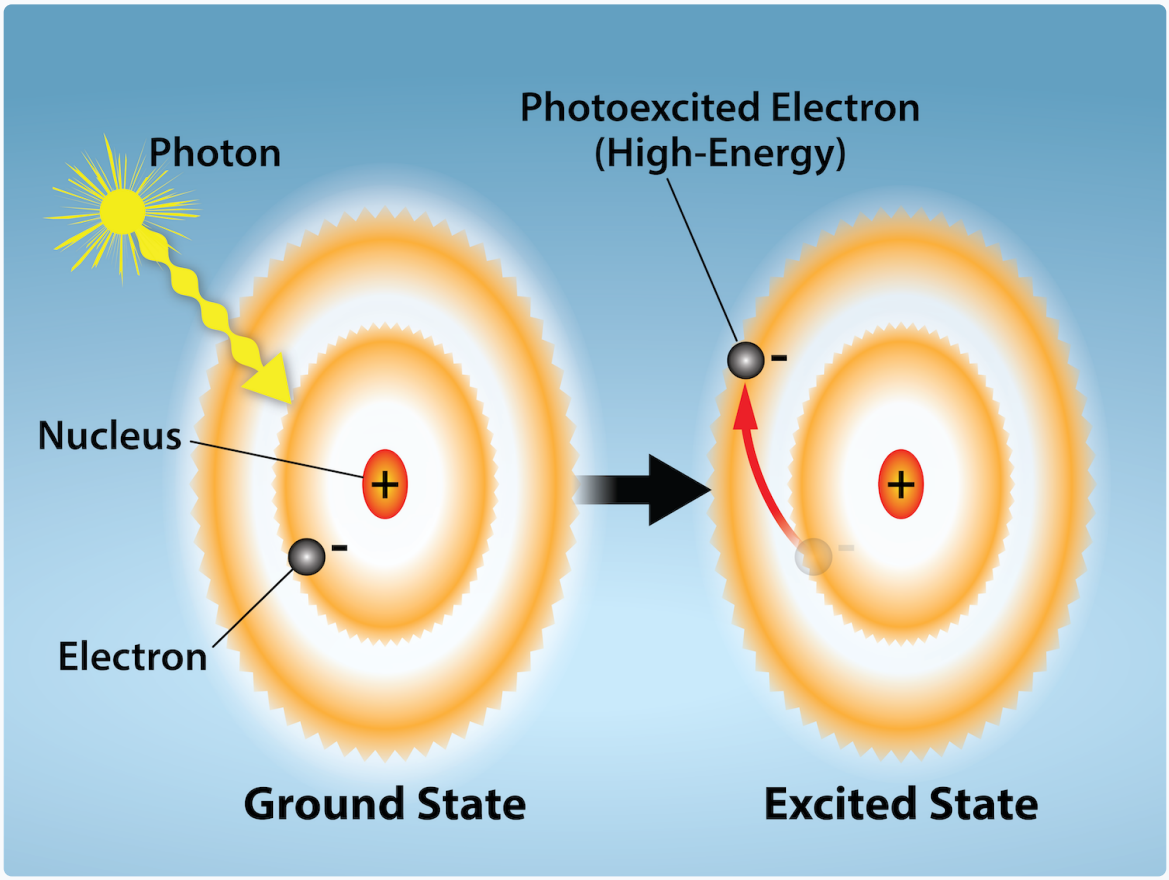
Figure 8.11 The sun emits energy in the form of electromagnetic radiation. This radiation exists at different wavelengths, each of which has its own characteristic energy. All electromagnetic radiation, including visible light, is characterized by its wavelength.

Each type of electromagnetic radiation travels at a particular wavelength. The longer the wavelength, the less energy it carries. Short, tight waves carry the most energy. This may seem illogical, but think of it in terms of a piece of moving heavy rope. It takes little effort by a person to move a rope in long, wide waves. To make a rope move in short, tight waves, a person would need to apply significantly more energy.

The electromagnetic spectrum (Figure 8.11) shows several types of electromagnetic radiation originating from the sun, including X-rays and ultraviolet (UV) rays. The higher-energy waves can penetrate tissues and damage cells and DNA, which explains why both X-rays and UV rays can be harmful to living organisms.

Absorption of Light

Light energy initiates the process of photosynthesis when pigments absorb specific wavelengths of visible light. Organic pigments, whether in the human retina or the chloroplast thylakoid, have a narrow range of energy levels that they can absorb. Energy levels lower than those represented by red light are insufficient to raise an orbital electron to an excited (quantum) state. Energy levels higher than those in blue light will physically tear the molecules apart, in a process called bleaching. Our retinal pigments can only “see” (absorb) wavelengths between 700 nm and 400 nm of light, a spectrum that is therefore called visible light. For the same reasons, plants, pigment molecules absorb only light in the wavelength range of 700 nm to 400 nm; plant physiologists refer to this range for plants as photosynthetically active radiation.



A ground state atom is depicted with an electron in an inner shell. A photon is pictured hitting the atom. The result is pictured as an excited state atom. The electron has moved to an outer shell.

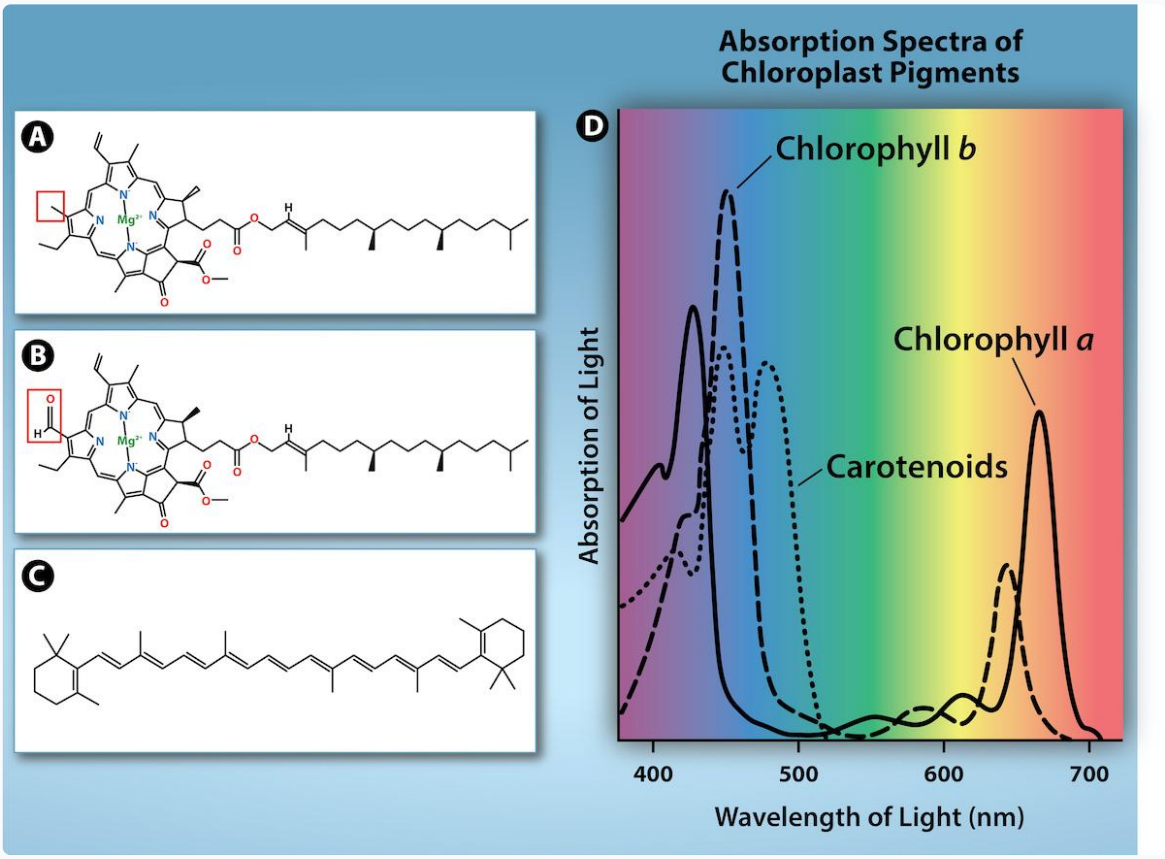
Figure 8.12 Light energy can excite electrons. When a photon of light energy interacts with an electron, the electron may absorb the energy and jump from its lowest energy ground state to an excited state. Credit: Rao, A. and Ryan, K. Department of Biology, Texas A&M University.

Understanding Pigments

Different kinds of pigments exist, and each absorbs only specific wavelengths (colors) of visible light. Pigments reflect or transmit the wavelengths they cannot absorb, making them appear a mixture of the reflected or transmitted light colors.

Chlorophylls and carotenoids are the two major classes of photosynthetic pigments found in plants and algae; each class has multiple types of pigment molecules. Chlorophyll *a* and chlorophyll *b* are found in higher plant chloroplasts and will be the focus of the following discussion.

Each type of pigment can be identified by the specific pattern of wavelengths it absorbs from visible light: This is termed the absorption spectrum. The graph in Figure 8.14 shows the absorption spectra for chlorophyll *a*, chlorophyll *b*, and a type of carotenoid pigment called β-carotene (which absorbs blue and green light). Notice how each pigment has a distinct set of peaks and troughs, revealing a highly specific pattern of absorption. Chlorophyll *a* absorbs wavelengths from either end of the visible spectrum (blue and red), but not green. Because green is reflected or transmitted, chlorophyll appears green. Carotenoids absorb in the short-wavelength blue region, and reflect the longer yellow, red, and orange wavelengths.



Chlorophyll a and chlorophyll b are made up of a long hydrocarbon chain attached to a large, complex ring made up of nitrogen and carbon. Magnesium is associated with the center of the ring. Chlorophyll b differs from chlorophyll a in that ...

Figure 8.14 (a) Chlorophyll a, (b) chlorophyll b, and (c) β-carotene are hydrophobic organic pigments found in the thylakoid membrane. Chlorophyll a and b, which are identical except for the part indicated in the red box, are responsible for the green color of leaves. β-carotene is responsible for the orange color in carrots. Each pigment has (d) a unique absorbance spectrum. Credit: Rao, A., Ryan, K., Tag, A., Fletcher, S. and Hawkins, A. Department of Biology, Texas A&M University.

Many photosynthetic organisms have a mixture of pigments, and by using these pigments, the organism can absorb energy from a wider range of wavelengths. Not all photosynthetic organisms have full access to sunlight. Some organisms grow underwater where light intensity and quality decrease and change with depth. Other organisms grow in competition for light. Plants on the rainforest floor must be able to absorb any bit of light that comes through, because the taller trees absorb most of the sunlight and scatter the remaining solar radiation (Figure 8.15).

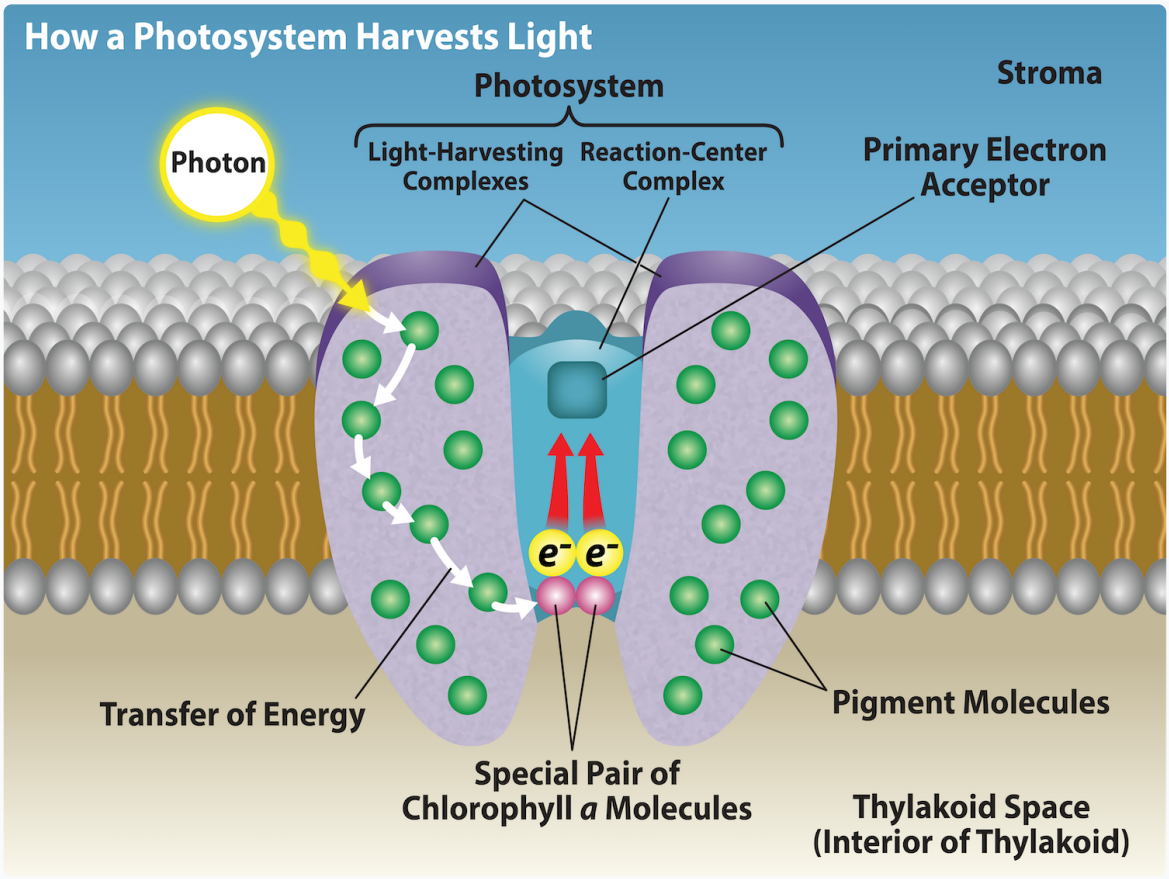


The photo shows undergrowth in a forest.

Figure 8.15 Plants that commonly grow in the shade have adapted to low levels of light by changing the relative concentrations of their chlorophyll pigments. (credit: Jason Hollinger)

How the Light Reaction (Light-Dependent Reaction) Works

The overall function of the Light Reaction is to convert solar energy into chemical energy in the form of NADPH and ATP. This chemical energy supports the Calvin Cycle and fuels the assembly of sugar molecules. The Light Reaction is depicted in Figure 8.16 . Protein complexes and pigment molecules work together to produce NADPH and ATP. The numbering of the photosystems is derived from the order in which they were discovered, not in the order of the transfer of electrons.

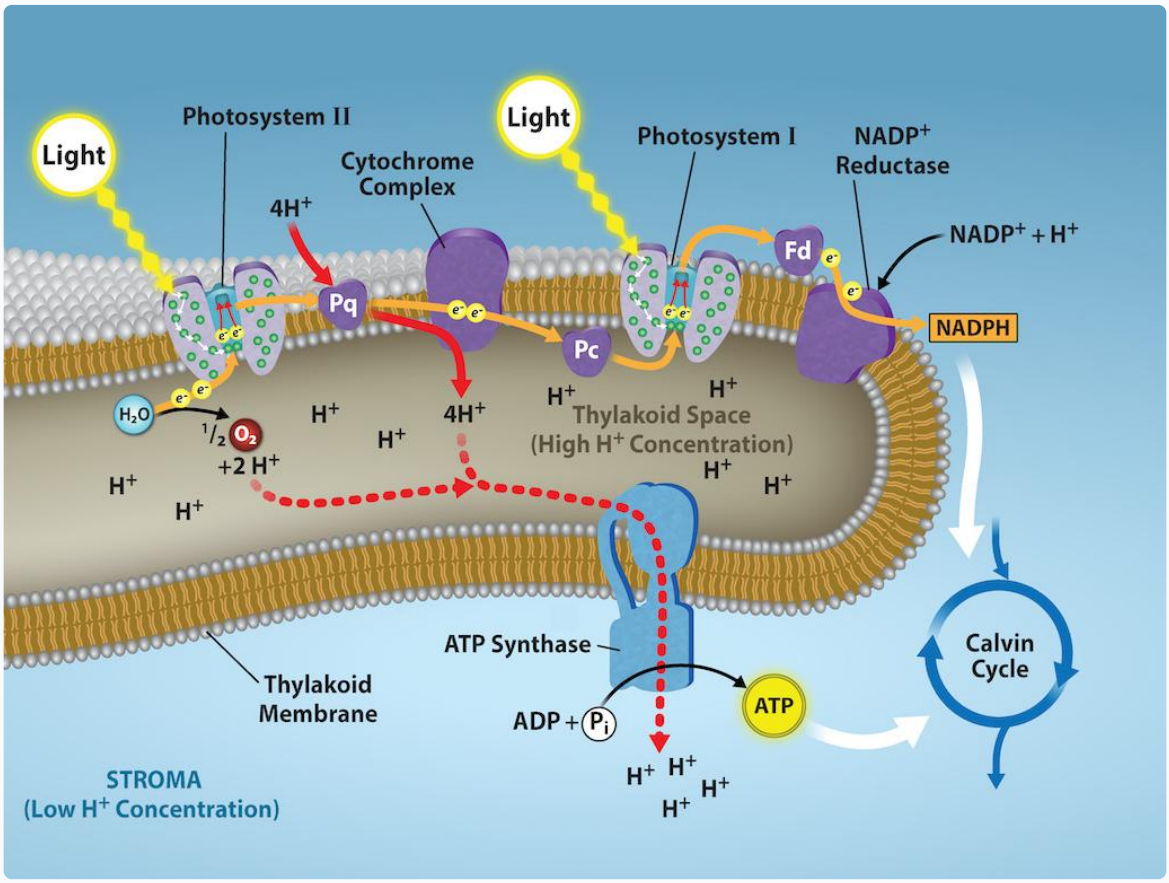


A photosystem is shown embedded in the thylakoid membrane. The reaction center is surrounded by the light-harvesting complexes, which contain pigment molecules. When a photon interacts with the pigment molecules, they transfer light energy ...

Figure 8.16 A photosystem consists of 1) a light-harvesting complex and 2) a reaction center. Pigments in the light-harvesting complex pass light energy to two special chlorophyll a molecules in the reaction center. The light excites an electron from the chlorophyll a pair, which passes to the primary electron acceptor. The excited electron must then be replaced. Credit: Rao, A., Ryan, K., Fletcher, S. and Hawkins, A. Department of Biology, Texas A&M University.

The actual step that converts light energy into chemical energy takes place in a multiprotein complex called a photosystem, two types of which are found embedded in the thylakoid membrane: Photosystem II (PSII) and Photosystem I (PSI) (Figure 8.17). Both photosystems have the same basic structure of chlorophyll molecules surrounding the reaction center where the photochemistry takes place. Each photosystem is serviced by the light-harvesting complex, which passes energy from sunlight to the reaction center. The absorption of a single photon or distinct quantity or "packet" of light by any of the chlorophylls pushes that molecule into an excited state. In short, the light energy has now been captured by biological molecules but is not stored in any useful form yet. The energy is transferred from chlorophyll to chlorophyll until eventually (after about a millionth of a second), it is delivered to the reaction center. Up to this point, only energy has been transferred between molecules, not electrons.

VISUAL CONNECTION



This illustration shows the components involved in the light reactions, which are all embedded in the thylakoid membrane. Photosystem I I uses light energy to strip electrons from water, producing half an oxygen molecule and two protons in ...

Figure 8.17 In the photosystem II (PSII) reaction center, energy from sunlight is used to extract electrons from water. The electrons travel through the chloroplast electron transport chain to photosystem I (PSI), which reduces $NADP^+$ to NADPH. The electron transport chain moves protons across the thylakoid membrane into the lumen. At the same time, splitting of water adds protons to the lumen, and reduction of NADPH removes protons from the stroma. The net result is a low pH in the thylakoid lumen, and a high pH in the stroma. ATP synthase uses this electrochemical gradient to make ATP. Credit: Rao, A., Ryan, K., Fletcher, S. Department of Biology, Texas A&M University.

The reaction center contains a pair of chlorophyll *a* molecules with a special property. Those two chlorophylls can undergo oxidation upon excitation; they can actually give up an electron. It is at this step in the reaction center during photosynthesis that light energy is converted into an excited electron. All of the subsequent steps involve getting that electron to the energy carrier NADPH for delivery to the Calvin cycle where the electron is deposited onto carbon for long-term storage in the form of a carbohydrate. Photosystem II and Photosystem I are two major components of the photosynthetic electron transport chain, which also includes the *cytochrome complex*. The cytochrome complex, an enzyme composed of two protein complexes, transfers the electrons from the carrier molecule plastoquinone (Pq) to the protein plastocyanin (Pc), thus enabling both the transfer of protons across the thylakoid membrane and the transfer of electrons from Photosystem II to Photosystem I.

The reaction center of Photosystem II (called P680) delivers its high-energy electrons, one at the time, to the primary electron acceptor, and through the electron transport chain (Pq to cytochrome complex to plastocyanine) to Photosystem I. Photosystem II's missing electron is replaced by extracting a low-energy

electron from water; thus, water is “split” during this stage of photosynthesis, and Photosystem II is reduced. Splitting one H_2O molecule releases two electrons, two hydrogen atoms, and one atom of oxygen. About 10 percent of the oxygen is used by mitochondria in the leaf to support oxidative phosphorylation. The remainder escapes to the atmosphere where it is used by aerobic organisms to support respiration.

As electrons move through the proteins that reside between Photosystem II and Photosystem I, they lose energy. This energy is used to move hydrogen atoms from the stroma, across the thylakoid membrane, into the thylakoid space. Those hydrogen atoms, plus the ones produced by splitting water, accumulate in the thylakoid space, creating a concentration gradient, and will be used synthesize ATP in a later step. Because the electrons have lost energy prior to their arrival at Photosystem I, they must be re-energized by another photon of light. That energy is relayed to the Photosystem I reaction center (called P700). P700 is oxidized and sends a high-energy electron to $NADP^+$ to form NADPH, an electron carrier. The two photosystems work in concert to produce NADPH and ATP, both of which will be used in the Calvin Cycle.

Generating an Energy Carrier: ATP

As in the intermembrane space of the mitochondria during cellular respiration, the buildup of hydrogen ions inside the thylakoid space creates a *concentration gradient*. The passive diffusion of hydrogen ions from high concentration (in the thylakoid space) to low concentration (in the stroma) is harnessed to create ATP, just as in the electron transport chain of cellular respiration. To release this energy, hydrogen ions will rush through any opening, similar to water jetting through a hole in a dam. In the thylakoid, that opening is a passage through a specialized protein channel called the ATP synthase. The energy released by the hydrogen ion stream allows ATP synthase to attach a third phosphate group to ADP, which forms a molecule of ATP (Figure 8.17). The flow of hydrogen ions through ATP synthase is called chemiosmosis because the ions move from an area of high to an area of low concentration through a semi-permeable structure of the thylakoid.

Section 10.3: The Calvin Cycle (Light-Independent Synthesis Phase)

Stroma carbon fixation via RuBisCO, reduction of 3-PGA to G3P, and regeneration of RuBP.

Chapter 10 The Calvin Cycle

Learning Objectives

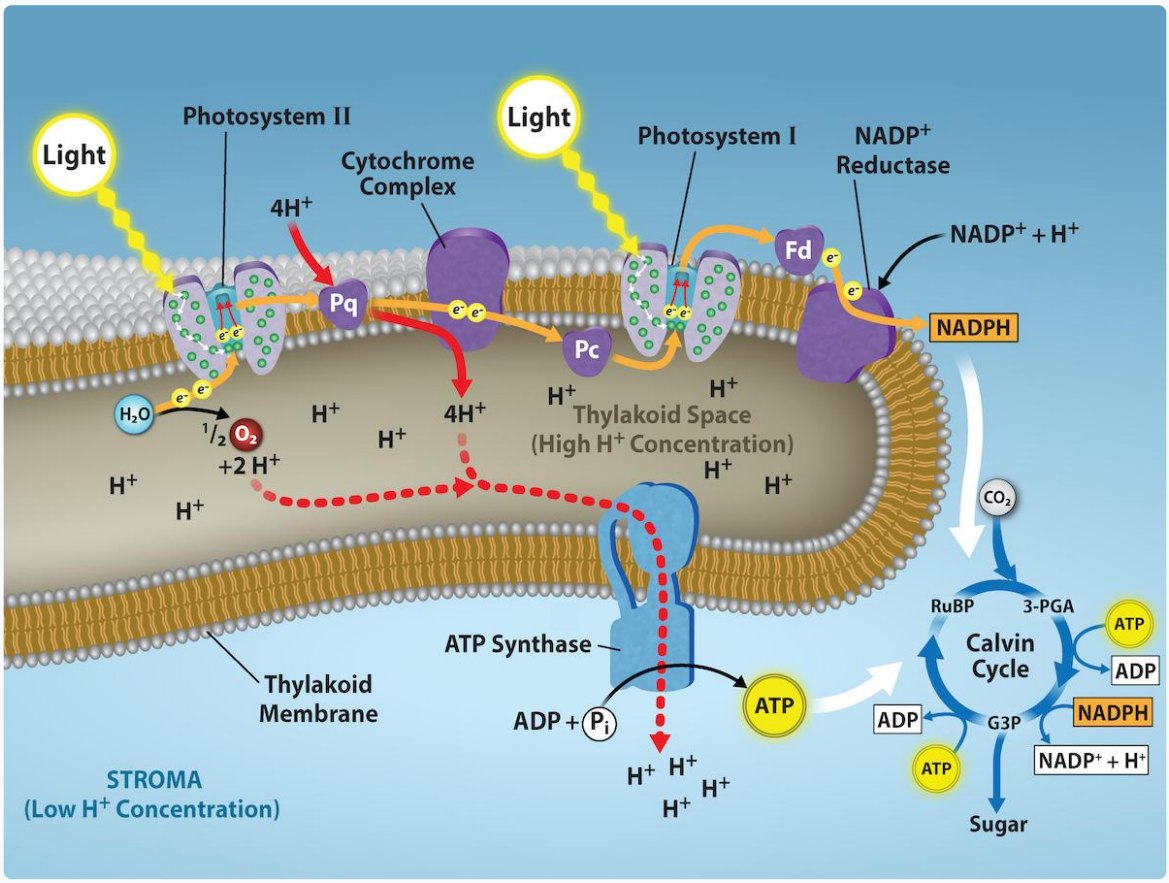
By the end of this section, you will be able to do the following:

- Describe the Calvin cycle
- Define carbon fixation

After the energy from the sun is converted into chemical energy and temporarily stored in ATP and NADPH molecules, the cell has the fuel needed to build carbohydrate molecules for long-term energy storage. The products of the Light Reaction, ATP and NADPH, have lifespans in the range of millionths of seconds, whereas the products of the Calvin Cycle, the light-independent cycle, G3P, can survive almost indefinitely. The carbohydrate molecules made will have a backbone of carbon atoms. But where does the carbon come from? It comes from carbon dioxide—the gas that is a waste product of respiration in microbes, fungi, plants, and animals.

The Calvin Cycle

In plants, carbon dioxide (CO_2) enters the leaves through stomata, where it diffuses over short distances through intercellular spaces until it reaches the mesophyll cells. Once in the mesophyll cells, CO_2 diffuses into the stroma of the chloroplast—the site of the Calvin Cycle. These reactions actually have several names associated with them.



This illustration shows that A T P and N A D P H produced in the light reactions are used in the Calvin cycle to make sugar.

Figure 8.18 Light reactions harness energy from the sun to produce chemical bonds, ATP, and NADPH. These energy-carrying molecules are made in the stroma where carbon fixation takes place. Credit: Rao, A., Ryan, K., Tag, A., Fletcher, S. and Hawkins, A. Department of Biology, Texas A&M University.

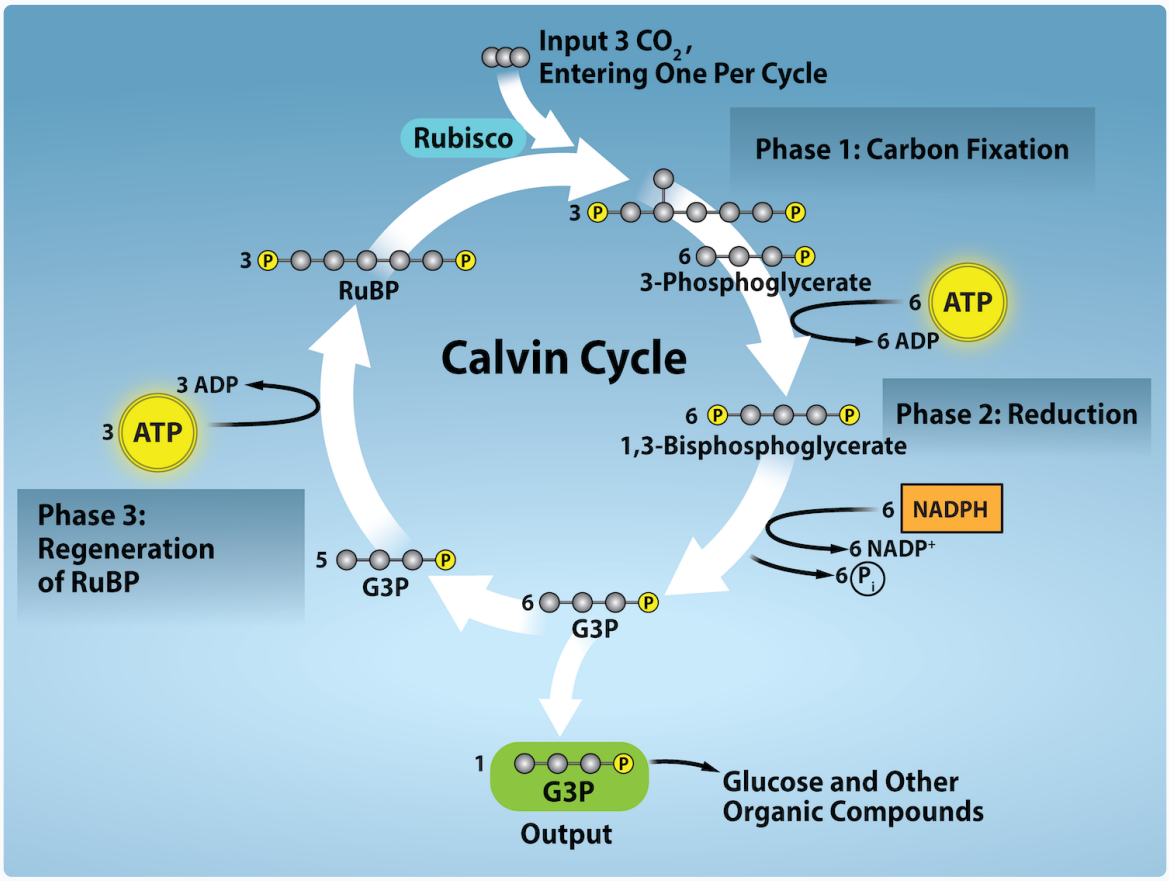
The light-independent reactions of the Calvin cycle can be organized into three basic stages: *fixation, reduction, and regeneration.*

Stage 1: Fixation

In the stroma, in addition to CO_2, two other components are present to initiate the light-independent reactions: an enzyme called ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO), and three molecules of ribulose bisphosphate (RuBP), as shown in Figure 8.19 . RuBP has five atoms of carbon, flanked by two phosphates.

VISUAL CONNECTION

Visual Connection



A diagram of the Calvin cycle is shown with its three stages: carbon fixation, 3 dash P G A reduction, and regeneration of upper case R lower case u upper case B upper case P. In stage 1, the enzyme upper R lower u upper B lower i lower s u...

Figure 8.19 The Calvin cycle has three stages. In stage 1, the enzyme RuBisCO incorporates carbon dioxide into an organic molecule, 3-PGA. In stage 2, the organic molecule is reduced using electrons supplied by NADPH. In stage 3, RuBP, the molecule that starts the cycle, is regenerated so that the cycle can continue. Only one carbon dioxide molecule is incorporated at a time, so the cycle must be completed three times to produce a single three-carbon G3P molecule, and six times to produce a six-carbon glucose molecule. Credit: Rao, A., Ryan, K., Tag, A., Fletcher, S. and Hawkins, A. Department of Biology, Texas A&M University.

RuBisCO catalyzes a reaction between CO_2 and RuBP. For each CO_2 molecule that reacts with one RuBP, two molecules of another compound 3-phospho glyceric acid (3-PGA) form. PGA has three carbons and one phosphate. Each turn of the cycle involves only one RuBP and one carbon dioxide and forms two molecules of 3-PGA. This fixation cycle "turns" three times such that: 3 carbon atoms from $3CO_2$ + 15 carbon atoms from 3 RuBP = 18 carbon atoms in 6 molecules of 3-PGA. This process is called **carbon fixation**, because CO_2 is "fixed" from an inorganic form into organic molecules.

Stage 2: Reduction

ATP and NADPH are used to convert the six molecules of 3-PGA into six molecules of glyceraldehyde 3-phosphate (**G3P**). That is a reduction reaction because it involves the gain of electrons by 3-PGA. Six molecules of ATP and 6 molecules of NADPH are required for this reduction stage.

Stage 3: Regeneration

Interestingly, at this point, only one of the G3P molecules leaves the Calvin cycle and is sent to the cytoplasm to contribute to the formation of glucose (two G3P molecules can be combined to create one glucose). After three "turns" of the Calvin Cycle, a total of 6 molecules of G3P have been created. One molecule is used by the plant to create glucose, while the remaining five G3P molecules remain in the cycle and are used to regenerate RuBP, which enables the system to prepare for more CO_2 to be fixed. Three more molecules of ATP are used in these regenerate the three molecules of RuBP.

LINK TO LEARNING

This link leads to an animation of photosynthesis and the Calvin cycle.

Section 10.4: Instructor Lecture Slide Review (27 Slides Synthesis)

Key concepts, equations, and diagrams from Dr. Dawson's Chapter 10 lecture presentation.

Slides 1–5: Course Reminders, Scope & The Photosynthesis Equation

Photosynthesis Equation: 6 CO_2 + 6 H_2O + Light Energy → $C_6H_{12}O_6$ + 6 O_2.

Thermodynamic Classification: Photosynthesis is an *anabolic* and *endergonic* process (+ΔG). In cellular respiration, glucose is catabolically oxidized to CO_2 and oxygen is reduced to water. In photosynthesis, water is oxidized to O_2 and CO_2 is anabolically reduced to sugar.

Slides 6–9: Chloroplast Ultrastructure & Chlorophyll Excitation

Chloroplast Compartmentalization: Double membrane (outer and inner membranes separated by an intermembrane space), liquid stroma interior, and flattened thylakoid sacs stacked into grana (singular: granum). The thylakoid membrane encloses the thylakoid lumen/space.

Chlorophyll Photon Excitation: When chlorophyll absorbs a photon of light, an electron jumps from its stable ground orbital to a higher energetic orbital (excited state). As the electron falls back, resonance energy is transmitted across adjacent pigment molecules until reaching the reaction center chlorophyll a pair.

Slides 10–18: The Light Reaction • Linear Electron Flow

Localization & Inputs: Occurs in the thylakoid membrane during daylight. Inputs: Light energy, H_2O, ADP, $NADP^+$. Outputs: O_2 (released to atmosphere), ATP (to Calvin cycle), NADPH (to Calvin cycle).

Two Photosystems in Series:

- **Photosystem II (P680):** Upstream photosystem. Absorbs 680 nm light; excited P680 transfers electrons to primary acceptor. Recharged by splitting water: $2\ H_2O \rightarrow 4\ H^+ + 4\ e^- + O_2$.
- **Electron Transport Chain:** Plastoquinone (Pq) → Cytochrome b_6f Complex → Plastocyanin (Pc). Pumping of H^+ into thylakoid lumen creates electrochemical proton gradient. ATP Synthase generates ATP in stroma via chemiosmosis.
- **Photosystem I (P700):** Absorbs 700 nm light; recharged by electrons from Pc. Transfers electrons via Ferredoxin (Fd) to $NADP^+$ Reductase, producing NADPH in the stroma.

Slides 19–27: The Calvin Cycle • 3-Phase Enzymatic Wheel

Localization & Function: Occurs in the stroma. Anabolic pathway converting inorganic CO_2 into 3-carbon sugar (G3P). Operates during day or night as long as ATP and NADPH are available.

Three Sub-Phases of the Calvin Cycle:

1. **Phase 1: Carbon Fixation:** The enzyme *RuBisCO* combines 3 CO_2 with 3 five-carbon molecules of Ribulose Bisphosphate (RuBP), forming unstable 6C intermediates that split into 6 molecules of 3-Phosphoglycerate (3-PGA).
2. **Phase 2: Reduction:** 6 ATP phosphorylate 3-PGA to 1,3-bisphosphoglycerate, and 6 NADPH reduce it to 6 molecules of Glyceraldehyde 3-Phosphate (G3P). Exactly **1 G3P** exits the cycle to form glucose, starch, and cellulose; the remaining 5 G3P continue in the cycle.
3. **Phase 3: Regeneration of RuBP:** 3 ATP are consumed to rearrange the 5 G3P (15 carbons) back into 3 molecules of RuBP (15 carbons), resetting the cycle.

Net Stoichiometry per 1 G3P Sugar: $3\ CO_2 + 9\ ATP + 6\ NADPH \rightarrow 1\ G3P + 9\ ADP + 8\ P_i + 6\ NADP^+$. Synthesis of 1 glucose (6C) requires 2 G3P (6 CO_2, 18 ATP, 12 NADPH).

Section 10.5: Official Course Practice Questions & Solutions

All 27 official examination problems from Chapter 10 Practice Questions.docx with verified answers.

1. Photosynthesis is a(n) _____ pathway & is therefore a _____ process.

A) Anabolic, exergonic • B) Catabolic, exergonic • **C) Anabolic, endergonic (Correct)** • D) Catabolic, endergonic

2. The light reaction takes place in the _____:

A) Thylakoid membrane (Correct) • B) Granum • C) Photosystem • D) Photon

3. T or F. Plants absorb green light waves used for photosynthesis:

A) True • **B) False (Correct: Green wavelengths are reflected, not absorbed)**

4. Which of the following occurs in the stroma of the chloroplast?

A) Light reaction • B) Electron transport chain • **C) Calvin cycle (Correct)** • D) Glycolysis

5. During what stage of photosynthesis is O_2 produced?

A) Cyclic photophosphorylation • **B) The light dependent reactions involving photosystem II & I (Correct: photolysis at PS II)** • C) Carbon fixation • D) The Krebs cycle

6. Which of the following is not a product of the Light Reaction?

A) Oxygen • B) ATP • C) NADPH • **D) G3P (Correct: G3P is the product of the Calvin cycle)**

7. The Calvin Cycle requires the input of each of the following except…

A) ATP • B) NADPH • **C) O_2 (Correct: O_2 is not required by Calvin cycle)** • D) CO_2

8. In the Light Reaction, when light strikes the chlorophyll pigments (P700 or P680) what is the immediate result?

A) Excited electrons are passed to electron acceptors (Correct) • B) Electrons are fused to form ATP • C) Glucose is produced • D) Carbon fixation occurs

9. Which color below is reflected off the thylakoid?

A) Red • B) Blue • C) Yellow • **D) Green (Correct)**

10. T or F. The Calvin Cycle is the light dependent reaction involved in photosynthesis:

A) True • **B) False (Correct: Calvin cycle is light-independent)**

11. Select the correct sequence for the movement of electrons during the light dependent reaction of plants: (I. Photosystem I, II. Photosystem II, III. Water, IV. $NADP^+$)

A) I, II, III, IV • B) I, IV, III, II • **C) III, II, I, IV (Water → PS II → PS I → $NADP^+$) (Correct)** • D) I, III, II, IV

12. During which stage of photosynthesis are ATP & NADPH converted to ADP + P_i & $NADP^+$?

A) Light dependent • **B) Calvin Cycle (Correct: consumes ATP and NADPH)** • C) Both of the above • D) None of the above

13. T or F. Light-driven electron transport in the thylakoid pumps H^+ into the intermembrane space between the outer & inner membranes:

A) True • **B) False (Correct: H^+ is pumped into the thylakoid lumen/space, NOT intermembrane space)**

14. The main purpose of photosynthesis is to:

A) Consume CO_2 • B) Produce ATP • **C) Produce G3P (sugar) (Correct)** • D) Convert light energy into kinetic energy

15. Where in the chloroplast is the H^+ ion concentration gradient built up?

A) In the stroma • **B) In the thylakoid space (Correct: creates low pH lumen)** • C) In the intermembrane space • D) In the cytosol

16. How is Photosystem II recharged?

A) Splitting water provides electrons to Photosystem II (Correct) • B) Oxygen provides electrons • C) Photosystem I provides electrons • D) $NADP^+$ provides electrons

17. How is Photosystem I recharged?

A) Splitting water provides electrons to PS II • B) Oxygen provides electrons • **C) Photosystem II provides electrons to Photosystem I via the ETC (Correct)** • D) $NADP^+$ provides electrons

18. As the Light Reaction progresses, the pH in the thylakoid space:

A) Increases (becomes more basic) • **B) Decreases (becomes more acidic as H^+ ions accumulate) (Correct)** • C) Stays the same

19. What is the final electron acceptor of the electrons from Photosystems II and I?

A) ATP • B) CO_2 • C) O_2 • **D) $NADP^+$ (Correct: forms NADPH)**

20. What happens to photosynthesis in the absence of sunlight?

A) The Light Reaction cannot take place (Correct) • B) The Calvin Cycle cannot take place immediately • C) NADPH and ATP continue to be produced • D) The plant switches to the Dark Reaction

21. The Light Reaction needs $NADP^+$ and ADP, where does it get these molecules?

A) Produced in chloroplast cytosol • B) Plant mitochondria produce them • **C) They are a byproduct of the Calvin Cycle (Correct: recycled back to thylakoids)** • D) Produced from splitting H_2O

22. _____ must be regenerated at the end of the Calvin Cycle:

A) G3P • **B) RuBP (Ribulose Bisphosphate) (Correct)** • C) ATP • D) CO_2

23. Which of the following statements is correct?

A) Plants use mitochondria for photosynthesis • **B) Plants produce waste oxygen that animals use in their mitochondria for cellular respiration (Correct)** • C) Animals produce waste oxygen • D) Animals and plants both produce waste oxygen

24. Photosystem II...

A) Uses light energy to transfer electrons to water • B) Takes electrons from water and passes to ATP Synthase • **C) Takes electrons from water, energizes them, and passes them to an electron transport chain (Correct)** • D) Absorbs light energy to energize electrons and make NADPH

25. Evaluate the accuracy of the statement: "ATP Synthase takes electrons from the electron transport chain and makes ATP."

A) Actually, ATP Synthase uses the H^+ gradient produced by the electron transport chain to make ATP, it does not use electrons themselves (Correct) • B) ATP Synthase takes electrons to break down ATP • C) ATP Synthase doesn't depend on ETC • D) Statement is accurate

26. Photosystem I...

A) Sends electrons to PS II • B) Makes NADPH used by ATP Synthase • C) Uses H^+ gradient to make ATP • **D) Absorbs light to energize electrons so they can be transferred to $NADP^+$ to make NADPH (Correct)**

27. Both ATP and NADPH are made in the stroma of the chloroplast. What would happen if they were trapped within the thylakoid space?

Answer: The Calvin Cycle would shut down because the enzymes of the Calvin cycle (such as RuBisCO and G3P dehydrogenases) reside in the stroma and require stroma-localized ATP and NADPH as energy and reducing power.

Appendix: Course Active Learning Practice Sets & Solutions

Original in-class problem sets, chloroplast concept mapping, and herbicide inhibition diagnostics from Active Learning #3 (Slides 8–9).

Problem 1: Chloroplast Concept Mapping (Active Learning #3, Slide 8)

Trace the path of energy, electrons, and carbon across chloroplast structural compartments:

Component	Localization	Functional Role & Mechanism
Photosystem II (P680)	Thylakoid Membrane	Absorbs light; extracts electrons via water splitting: $2\ H_2O \rightarrow 4\ H^+ + 4\ e^- + O_2$.
Cytochrome b_6f Complex	Thylakoid Membrane	Transfers electrons from Pq to Pc; pumps H^+ from stroma into thylakoid lumen to generate proton gradient.
Photosystem I (P700)	Thylakoid Membrane	Re-excites electrons using light energy; passes electrons via ferredoxin to $NADP^+$ reductase to generate NADPH.
ATP Synthase	Thylakoid Membrane	Drives photophosphorylation as H^+ ions flow down their electrochemical gradient from lumen back into the stroma.
Calvin Cycle	Chloroplast Stroma	Light-independent phase: RuBisCO fixes CO_2; consumes 9 ATP and 6 NADPH to yield 1 net G3P per 3 CO_2.

Problem 2: Herbicides Inhibition Analysis (Active Learning #3, Slide 9)

Five distinct experimental herbicides (H-1 through H-5) are applied to isolated chloroplasts. Based on physiological readouts, deduce the specific molecular target of each compound:

Herbicide	Chlorophyll Absorption	O_2 Output	NADPH Output	ATP Synthase	G3P Output	Molecular Target & Diagnostic Rationale
H-1	Normal	Normal	**Decreased**	Normal	**Decreased**	**Photosystem I:** PS II splits water normally (O_2 normal) and proton pumping drives ATP synthesis normally, but electrons cannot reduce $NADP^+$ to NADPH. Without NADPH, Calvin cycle stalls (G3P decreases).
H-2	**Decreased**	**Decreased**	**Decreased**	**Decreased**	**Decreased**	**Chlorophyll Absorption:** Primary photon harvesting is blocked. Without light excitation, water splitting, proton pumping, photophosphorylation, and carbon fixation all shut down.
H-3	Normal	**Decreased**	**Decreased**	**Decreased**	**Decreased**	**Photosystem II:** PS II water oxidation is halted, dropping O_2. Without electron donation from PS II, downstream electron transport, proton gradient generation, NADPH, and G3P crash.
H-4	Normal	Normal	Normal	Normal	**Decreased**	**Calvin Cycle (e.g., RuBisCO):** All thylakoid light reactions run normally (light absorption, O_2 evolution, ATP, and NADPH intact), but carbon fixation is selectively blocked.
H-5	Normal	Normal	Normal	**Decreased**	**Decreased**	**ATP Synthase:** Light absorption, water splitting, and NADPH synthesis proceed unaffected, but photophosphorylation is blocked. Starved of ATP, the Calvin cycle halts (G3P decreases).